



QS-PID Complete Setup Guide

Prerequisites

- Node.js (v18+) installed
 - npm (comes with Node.js)
-

PHASE 1: Verify Dependencies & Configuration

Step 1: Check Node.js Installation

```
node --version  
# Should show v18.x.x or higher
```

Step 2: Check Project Structure

```
ls -la  
# You should see: frontend/, src/, circuits/, tests/, package.json
```

Step 3: Install Dependencies

```
npm install
```

PHASE 2: Backend Services

Option A: Frontend Only (Demo Mode) - RECOMMENDED FOR TESTING

This runs the web server with mock/demo proofs (no circuit compilation needed):

```
npm run frontend
```

Expected Output:

```
[✓] QS-PID frontend running at http://localhost:3000  
[*] Artifacts ready : false  
[*] Press Ctrl+C to stop
```

Option B: Full Backend (Real ZKP)

If you want actual cryptographic proofs:

```
# Compile Circom circuits
npm run compile

# Setup proving keys
npm run setup

# Then start frontend
npm run frontend
```

PHASE 3: Verify Services Running

Check Backend Status

1. Open browser to <http://localhost:3000>
2. Look at the **status bar** at the bottom of the hero section
3. You should see:
 - **Backend: ONLINE** (green dot) or OFFLINE (red dot)
 - **Artifacts Ready: NO** (in demo mode) or YES (if compiled)
 - **Node: v22.x.x**
 - **Uptime: Xs**

PHASE 4: Frontend Navigation

Page 1: Main Hub (<http://localhost:3000>)

What you see:

- Animated 3D wireframe (Three.js) - rotating icosahedron
- "QS-PID" title with "QUANTUM-SAFE · ZERO-KNOWLEDGE · POST-QUANTUM" badge
- Two buttons: " ⚡ Launch Demo" and " 📖 View Docs"
- 7 clickable cards (section cards)

Interactive elements:

- Scroll down to see all 7 cards
- Click any card to open in new tab
- Status bar shows backend health

Page 2: Generate Income Proof ([prove.html](#))

Location: Click first card " ⚡ Generate Income Proof"

Input Fields:

Field	Location	Example Value
Annual Income (LPA)	Top input	12
Threshold (LPA)	Second input	5
Verifier ID	Third input	demo-verifier-001

What happens:

- As you type in "Annual Income", the ₹ badge updates (e.g., 12 → ₹12,00,000)
- Same for Threshold

Button: " ⚡ Generate Proof"

- Click to generate ZKP
- Expected result:** Shows "✅ Income ABOVE threshold" (green) or "❌ Income BELOW threshold" (red)
- Privacy preserved: shows "Actual amount hidden"
- Mode: Shows "Demo" if no circuits compiled

Action Buttons (appear after generation):

- "📄 Copy Proof JSON" - copies to clipboard
- "✅ Verify This Proof" - opens verify.html with proof pre-filled

Page 3: Verify Proof (verify.html)

Location: Click "✅ Verify Proof" card OR from prove.html

Input Field:

Field	Location	What to Enter
Proof JSON	Large textarea	Paste proof JSON from prove page

Auto-fill: If you came from prove.html, the proof is auto-filled in sessionStorage

Button: "✅ Verify Proof"

- Click to verify
- Expected result:**
 - Green banner: "✅ PROOF VALID"
 - Red banner: "❌ PROOF INVALID"

Page 4: Post-Quantum ML-DSA (pq.html)

Location: Click "🔗 Post-Quantum ML-DSA" card

What you see:

- Info banner about NIST FIPS 204 ML-DSA-65

Button: "🔑 Generate ML-DSA-65 Key Pair"

- Click to generate key pair
 - **Expected result:**
 - Variant: ML-DSA-65
 - Created At: [timestamp]
 - Phase: Active
 - Public Key: (truncated hex)
 - "View Full JSON" expandable section
-

Page 5: System Status (status.html)

Location: Click " System Status" card

What you see (auto-loads on page open):

- 4 stat cards in 2x2 grid:
 - Backend Status: ONLINE/OFFLINE
 - Artifacts Ready: YES/NO
 - Node Version: v22.x.x
 - Uptime: Xs
- API Endpoints table
- " Refresh Status" button

Button: " Refresh Status"

- Click to reload status data
-

Page 6: Issuer Registry Dashboard (dashboard.html)

Location: Click " Issuer Registry" card

What you see:

- Info pill: "NIST FIPS 204: ML-DSA-65 ACTIVE · ZKP Curve: BN254 (Groth16)"
- 4 stat cards:
 - Issued Credentials: 1,248
 - Circuit Constraints: ~145K
 - Avg. Proof Time: 220 ms
 - Jaccard Similarity: < 0.05
- Table with 3 sample holders

Sample Data Explanation:

- These are DEMO/SAMPLE records showing the data structure
 - In production, this would connect to a real database
 - Columns marked PRIVATE are hidden from verifiers
 - Only Commitment (PUBLIC) can be shared
-

Page 7: Issue Credential (issue.html)

Location: Click " Issue Credential" card

Input Fields:

Field	Location	Example
Full Name	First input	Rahul Pal
Annual Income (INR)	Second input	750000
Employer	Dropdown	HDFC Bank
Issuance Date	Date picker	2026-03-07

Cryptographic Pipeline (non-editable, shows what's happening):

- ☒ 32-byte Salt - Randomness for security
- ☒ HybridSigner (ECDSA+ML-DSA) - Quantum-safe signatures
- ☒ JSON-LD Format - W3C standard

Buttons:

Button	Action
✓ Issue Credential	Generates the VC JSON
 Download	Downloads credential-{timestamp}.json
 Copy	Copies JSON to clipboard
 Load into Wallet	Opens wallet.html with credential loaded

Live Preview: The W3C VC 2.0 JSON updates as you type

Page 8: Wallet (wallet.html)

Location: Click " Wallet" card OR from issue.html via "Load into Wallet"

What you see (3 sections):

Section 1: Personal Info (PRIVATE)

- Name, Employer, Income (hidden from verifiers)

Section 2: Crypto Artifacts (ZK-READY)

- Poseidon Commitment
- Session Nonce
- ML-DSA-65 Signature

Button: " **Download Wallet**" - Downloads wallet backup

Section 3: ZKP Circuit Logic

Shows the 3 circuit steps:

- 1. ✓ Num2Bits(32) - Prevents overflow
- 2. ✓ Income > Threshold - Comparison
- 3. ✓ Fiat-Shamir - Challenge binding

Section 4: 🔗 Verifier Challenge (IMPORTANT!)

Purpose: Generate a proof showing "income > 5 LPA" WITHOUT revealing exact salary

Input Field:

Field	Location	Example
Challenge (optional)	Text input	bank-loan-2026

Button: "✓ Generate Proof"

- Click to create ZK proof
- **Expected result:** Green "☑ ZK Proof Generated" section appears
- Shows the proof JSON with privacy guarantee

Proof Output Actions:

- ⬇ Download - Saves proof-{timestamp}.json
- 📋 Copy - Copies to clipboard

How to use the proof:

- 1. Download or copy the generated proof
- 2. Send to verifier (bank, lender, etc.)
- 3. They can verify it shows "income > 5 LPA" without knowing exact amount

PHASE 5: Complete Workflow Example

Scenario: Apply for Loan

- 1. **Go to Issue Credential** → Fill details → Click "Issue Credential" → Click "Load into Wallet"
- 2. **In Wallet page:**
 - Your info is now loaded
 - Scroll to "🔗 Verifier Challenge"
 - Enter challenge: "loan-application-123"
 - Click "✓ Generate Proof"
- 3. **Download proof** (or copy)
- 4. **Send to bank** - They verify and only learn "income > 5 LPA" not your exact ₹7.5L

Troubleshooting

Issue: "Backend: OFFLINE"

- Make sure `npm run frontend` is running
- Check port 3000 isn't blocked

Issue: Network Error on prove/verify

- Server might be crashed - restart with `npm run frontend`

Issue: Favicon not showing

- Clear browser cache or use incognito mode
- Some browsers need explicit favicon.ico file

Issue: Three.js animation slow

- This is normal on mobile - it will be smoother on desktop
- The canvas has a pixel ratio cap for performance